

Automation & Workflows
DATA 440

Instructor Info
Dr. Ron Smith
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Office hours: T/R Noon-2pm, ISC 3378 (no office hours Aug 25)

Course Overview, Fall 2026
Section 13: MWF 11-11:50am, ISC 3342
Section 14: MWF 12-12:50pm, ISC 3342

Automation: Minimizing the amount of manual work required to reproduce the end result of a **workflow**: The individual steps required to complete a larger task.

Real-world analyses are rarely done just once. When analyzing data for a research publication, mistakes may be made, new data may become available, or someone else may want to reproduce all or part of your analysis. Many businesses rely on regular weekly, monthly or quarterly reports. Some applications may require analysis of a continuous feed of data. You might also find yourself working with a team of analysts/programmers. Many of these cases can benefit from automation and well-documented workflows. Developing such workflows often requires a good amount of thought and careful planning in order to achieve a good balance of abstraction and generalization without making the logic or usage of the application unnecessarily complex.

In this course, students will work on several projects with a focus on reproducibility, efficiency, and ease of maintenance and use, especially when it comes time for an analysis to be revised or redone. Examples will be drawn from real-world or synthetic data. Students are encouraged to suggest their own examples and bring their own data. Along the way, students will provide prototypes of their programs and documentation for other students to test and critique. At the end of the semester, students will present a project that they have designed along with written documentation and a working demonstration.

COLL 400 Designation: This is a COLL 400 course, which:

> *...will require students to take initiative in synthesis and critical analysis, to solve problems in an applied and/or academic setting, to create original material or original scholarship, and to communicate effectively with a diversity of audiences.*

In this capstone course, you will synthesize and apply what you have learned from the Data Science curriculum as well as the broader COLL curriculum to:

- Develop your own solution to a problem that you will define (perhaps with my guidance!)
- Assist others in solving their problems by testing their code, reading their documentation, and providing feedback.
- Build a library of code along with supporting documentation so that a user without detailed knowledge of how the code works can still use it.
- Present your work to the class, keeping your presentation at a level appropriate for a general (i.e., non-expert) audience.

Prerequisites
DATA 301 (Applied Machine Learning) is a required prerequisite.

As this is a project-based course that will require both working with data as well as presenting results, DATA 302 (Databases) and DATA 303 (Data Visualization) are recommended, but not currently required.

Course Structure: Part 1
For the first part of the semester you will work in small groups on a project that the professor will provide. Along the way, we will focus on topics such as:

- **GitHub**: Learn how to effectively use GitHub for project management, version control, and code sharing.
- **Virtual Environments**: Learn how to isolate the code and libraries for separate projects and minimize, as much as possible, the issues that can arise from code dependency issues or version conflict issues. This is to both safeguard yourself from potential issues and to make your code more readily usable by others.
- **Modular and Object-Oriented Programming**: Learn how to write code that can withstand the tests of time! Without delving too deep into the theory, we will discuss good design principles that you should be employing at all times for practical reasons, such as making sure the code is readable and easily maintainable, has a reduced risk of bugs creeping in, and will not require an entire rewrite when you have to change something. This applies to all the code you generate, whether you write it all yourself or use AI assistance.
- **Data Acquisition/Preprocessing/Engineering**: It is very uncommon that the data needed for a project will be presented/acquired in a ready-to-go state. Data preprocessing and engineering tasks are not just common; they can often require as much (if not more) work as the rest of the project.

- **Planning and Managing a Code-Based Project**: Most real-world analyses are never done just once. Additionally, as a project progresses, requirements may change, flaws in the data or logic might be discovered, or other new insights may be revealed that necessitate modification/revision. In combination with the topics above, we will focus on planning a project from inception to delivery, while also planning for the unknown but inevitable future where modifications will need to be made.
- **Documentation, Peer Review, Presentation**: Your code will be well-documented. You will review others' code and documentation. You will receive and provide feedback to each other about both the code and the clarity of the documentation. You will present your work to a non-expert audience in an appropriately accessible manner. When working on code, always keep the following principle in the forefront of your mind: **Readability counts**.

Course Structure: Part 2

For the second half of the semester you will work in small groups on a final project. This project will consist of specifying a problem of your choosing (with professor approval), finding a data source, designing and automating a workflow to solve the problem, publishing your work and documentation on GitHub, and presenting your results to the class.

Class time will be split between working on your projects and lectures/demos based on the collective needs of the class.

Your final project will be due at the time published by the university registrar.

Grading

- **Checkpoints**: There will be 4 checkpoints assigned throughout the semester which will total 40% of your grade. These checkpoints may consist of submitting code for review, submitting documentation for review, reviewing code/documentation from other students, or identifying problems with code I provide you.
- **Midterm / Project Proposals**: For your midterm, you will submit a project proposal for your final project. This will be worth 15% of your grade, and details will be discussed when we get closer to that time.
- **Presentations**: You will give short presentations to the class with updates on your project. There will be either 2 or 3 of these, the last of which will be at the end of the semester when your project is close to completion. Collectively, these will be worth 15% of your grade.
- **Final project**: Your final project will be published on GitHub, reviewed by the professor, and worth 30% of your grade.

Final Grades

Final grades will be assigned on a 10-point scale, e.g. 90-100 = A, 80-89 = B, etc. + and -

grades will only be assigned in the event I feel your grade should be bumped up a little.

Academic Integrity
You are free to use all resources available to you, including the internet, AI, and asking others for help.

You are the sole owner of your work. This applies to both individual and group work. If you are unable to explain work you have submitted, code you have written, or results you have obtained, this will be a very serious cause for concern.

Things you should **always** do:

- Make sure all code you submit is thoroughly documented/commented, clearly demonstrating that you understand it.
- Make sure you properly cite sources. This includes details such as work you collaborated on with others, use of AI, and URLs you used to obtain code or code samples.

Things you should **never** do:

- Copy code or writing from another person
- Blindly use code obtained from the internet or AI without any attempt to understand it
- Fail to acknowledge or cite sources, such as who you worked with, or what online/AI resources you used
- Lie, cheat, or steal, in any way, shape, or form

Important dates

Add/drop deadline: September 4
Withdrawal period begins: September 5
Labor Day (no class): September 7
Fall break: October 8-11
Withdrawal deadline: October 26
Thanksgiving break: November 25-29

Accommodations
It is the policy of William & Mary to accommodate students with disabilities and qualifying diagnosed conditions in accordance with federal and state laws. Any student who feels s/he may need an accommodation based on the impact of a learning, psychiatric, physical or chronic health diagnosis should be referred to Student Accessibility Services staff at 757-221-2509 or at sas@wm.edu. SAS staff will work with you to determine if accommodations are warranted, and if so, to help you obtain an official letter of accommodation.

Mental and Physical Well-Being
William & Mary recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available at W&M to help students navigate emotional/psychological, physical/medical, material/accessibility concerns, including:

- The W&M Counseling Center at (757) 221-3620. Services are free and confidential.
- The W&M Health Center at (757) 221-4386
- For additional support, resources, or questions, Contact the Dean of Students at 757-221-2510 or deanofstudents@wm.edu

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Student Success
Student Success supports students' personal growth, development, engagement, and belonging by providing holistic guidance as they navigate their own W&M journey. It encompasses the offices of Academic Wellbeing, Care Support Services, and Student Accessibility Services.

- For academic support such as tutoring, time management, study skills, and academic coaching, please contact Academic Wellbeing at wm.edu/academicwellbeing (academicwellbeing@wm.edu)
- For concerns about the wellbeing of a member of the William & Mary community or to seek assistance for interpersonal, academic, and wellness challenges, please contact Care Support Services at wm.edu/care (care@wm.edu)
- For accommodation needs or questions, please contact Student Accessibility Services at wm.edu/sas (sas@wm.edu)